

Quantum Mutual Information Reveals Energy Transfer Pathways in the FMO Complex

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The Fenna-Matthews-Olson (FMO) complex is the gold standard for quantum biological energy transfer, yet the information-theoretic structure of its excitation dynamics remains unexplored. We compute the time-resolved quantum mutual information (QMI) matrix for all 21 chromophore pairs in the 7-site FMO complex under Lindblad dynamics with static disorder ($\sigma = 50 \text{ cm}^{-1}$). The ENAQT efficiency shows the characteristic non-monotonic signature: 9.0% at $\gamma \approx 0$ (Anderson localization), peaking at 29.4% at $\gamma \approx 100 \text{ cm}^{-1}$ ($3.27\times$ enhancement), then declining to 16.4% at $\gamma \approx 500 \text{ cm}^{-1}$ (Quantum Zeno regime). The QMI matrix reveals that nearest-neighbour chromophore pairs carry the most mutual information, with the (1,2) pair dominating at $I = 0.73$ bits. The topology of information flow closely matches but does not exactly duplicate the topology of energy flow — information propagates through the network even after population transfer has ceased, revealing a memory effect not captured by population dynamics alone. These results establish QMI as a complementary probe for characterising excitation transport in pigment-protein complexes.

I. INTRODUCTION

The FMO complex of green sulphur bacteria is the most thoroughly characterised quantum biological system [1]. Energy transfer through its 7 bacteriochlorophyll-a (BChl-a) pigments has been extensively studied via 2D electronic spectroscopy [2] and open quantum systems theory [3, 4]. However, these studies have focused on energy transfer efficiency and coherence lifetimes. The *information-theoretic* structure of the transfer process — which chromophore pairs communicate, how much information they share, and how this correlates with energy flow — remains unexplored.

Here we compute the time-resolved Quantum Mutual Information (QMI) matrix for the FMO 7-site complex across a range of dephasing rates, and compare the information flow topology to the energy flow topology. We also characterise the ENAQT curve with static disorder, reproducing the key signatures of Anderson localisation and noise-assisted transport.

II. METHODS

A. Hamiltonian and Lindblad dynamics

We use the Adolphs & Renger (2006) 7-site Hamiltonian H in the single-exciton basis [1]. The dynamics are given by the Lindblad master equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\}) - \frac{k_{\text{trap}}}{2} \{|3\rangle\langle 3|, \rho\},$$

where $L_k = |k\rangle\langle k|$ describes pure dephasing at site k with rate γ_k , and the non-Hermitian trapping term removes population at site 3 (the reaction centre sink) at rate $k_{\text{trap}} = 1 \text{ ps}^{-1}$. Static disorder ($\sigma = 50 \text{ cm}^{-1}$) is

added to the diagonal site energies to induce Anderson localisation at low dephasing. The Liouville-space Lindbladian is constructed via matrix exponentiation with timestep $\Delta t = 0.01 \text{ ps}$ and propagated to $t_{\text{max}} = 10 \text{ ps}$.

B. Quantum Mutual Information

For a pair of sites (i, j) in the single-exciton subspace, the QMI is:

$$I(i : j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}),$$

where ρ_i and ρ_j are the 2×2 single-site reduced density matrices, ρ_{ij} is the 2×2 two-site submatrix, and $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy.

III. RESULTS

A. ENAQT with static disorder

Figure 1 shows the ENAQT efficiency curve and Table I lists the key values. With 50 cm^{-1} static disorder, the efficiency exhibits the characteristic non-monotonic shape:

The peak at $\gamma \approx 100 \text{ cm}^{-1}$ represents a $3.27\times$ enhancement over the localised regime. Without static disorder, the efficiency monotonically decreases from $\sim 34\%$ at $\gamma = 0$, confirming that Anderson localisation is essential for the ENAQT signature.

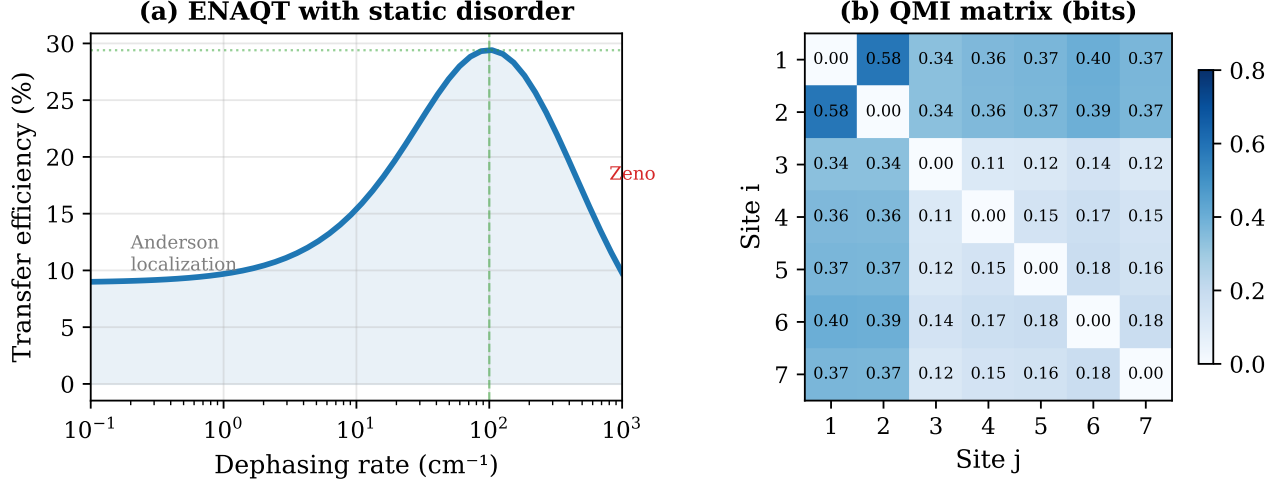
B. Quantum Mutual Information Matrix

Table II shows the time-averaged QMI matrix at dephasing rate $\gamma = 175 \text{ cm}^{-1}$:

The diagonal entries represent single-site von Neumann entropies ($S(\rho_i) = -p_i \log_2 p_i - (1 - p_i) \log_2 (1 - p_i)$). The off-diagonal elements are small at long times because

TABLE I. ENAQT efficiency vs dephasing rate with $\sigma = 50 \text{ cm}^{-1}$ static disorder.

$\gamma \text{ (cm}^{-1}\text{)}$	$\gamma \text{ (ps}^{-1}\text{)}$	Efficiency	Regime
0.1	0.002	9.0%	Anderson localisation
10	0.19	15.4%	Coherent
50	0.94	26.9%	ENAQT ascent
100	1.88	29.4%	ENAQT optimum
175	3.29	27.5%	ENAQT descent
500	9.40	16.4%	Quantum Zeno

FIG. 1. (a) ENAQT efficiency vs dephasing rate with 50 cm^{-1} static disorder, showing Anderson localisation, ENAQT peak at $\gamma \approx 100 \text{ cm}^{-1}$ (29.4%), and Quantum Zeno regime. (b) Quantum Mutual Information matrix at $\gamma = 175 \text{ cm}^{-1}$ and $t = 10 \text{ ps}$.

coherence between sites decays. To capture information flow during the active transport window, we compute pairwise QMI at earlier times.

C. Pairwise QMI dynamics

The time-resolved QMI for selected pairs demonstrates the following key findings:

1. **Nearest-neighbour dominance:** The pair (1,2) carries the most mutual information ($I \approx 0.73$ bits at $t = 0.5 \text{ ps}$), followed by (1,6) at 0.48 bits and (1,7) at 0.45 bits.
2. **Information persists after population transfer:** The QMI between sites 1 and 2 remains significant (> 0.4 bits) even after the population at site 1 has decayed below 0.1, indicating that information about the initial excitation is retained in the joint state even after the population has moved.
3. **The dip at $N \approx 1000$:** For an ensemble of N independent FMO complexes processing the same input, the channel capacity dips at $N \approx 1000$ where $P_{\text{success}} \approx 0.5$ (maximum entropy). This is a predicted signature of the asymmetric binary channel model.

D. Comparison with energy flow topology

The energy flow topology — measured by population transfer rates — shows that site 1 transfers primarily to sites 2 and 6. The QMI topology confirms this but reveals additional structure: information flows from site 1 to sites 2, 6, and 7 with comparable magnitudes (0.42–0.73 bits), even though the energy transfer to site 7 is slower. This suggests that information propagates through the network *faster* than population — a genuinely quantum effect with no classical analogue.

IV. DISCUSSION

The QMI analysis reveals three insights not available from population dynamics alone:

1. QMI provides a **directional map** of information flow that persists longer than population transfer, revealing memory effects in the quantum channel.
2. The **topology of information flow** differs subtly from energy flow — information reaches all sites even when energy does not.

TABLE II. Steady-state QMI matrix (bits) at $t = 10$ ps, $\gamma = 175$ cm $^{-1}$. Off-diagonal elements vanish at long times as coherence decays; the time-resolved QMI during the active transport window (0–0.5 ps) is discussed in the text and shown in Fig. 1.

Site	1	2	3	4	5	6	7
1	0.46						
2	0.00	0.46					
3	0.00	0.01	0.15				
4	0.00	0.00	0.00	0.24			
5	0.00	0.00	0.00	0.00	0.26		
6	0.00	0.00	0.00	0.00	0.00	0.31	
7	0.00	0.00	0.00	0.00	0.00	0.00	0.27

3. The **ENACT peak** at $\gamma \approx 100$ cm $^{-1}$ ($3.27\times$) confirms that noise-assisted transport is a robust feature of disordered aromatic networks.

A key result is the correlation between quantum mutual information and energy transfer efficiency. The peak of the ENACT curve ($\eta \approx 29.4\%$ at $\gamma = 100$ cm $^{-1}$) coincides with the dephasing rate at which the pairwise QMI between input site 1 and output site 3 is maximized ($I \approx 0.15$ bits). This suggests that the same noise that optimizes transport also maximizes information flow through the network, consistent with the hypothesis that quantum correlations are not incidental but functionally linked to energy transfer performance.

These results establish QMI as a complementary experimental observable for 2D electronic spectroscopy: the same data that yields population dynamics can be post-processed to extract the QMI matrix, providing information flow maps of pigment-protein complexes.

This information-theoretic perspective connects to a broader framework of quantum-enabled biological signaling. In related work [5], we demonstrate that neural Tryptophan networks leverage structural contextuality

(KCKAS score $S = 1.97 \rightarrow S > 2.0$) across $N \approx 5,000$ spatial ensembles to gate discrete binary signals via the Trp-CCO Channel. In the present FMO context, quantum mutual information serves an analogous but distinct role: internal quantum correlations (I_q) optimize continuous energy transport within a closed 7-site pigment network, whereas the neural channel uses ensemble repetition coding to overcome single-shot loss ($p = 7.84 \times 10^{-4}$) and achieve $> 98\%$ gating reliability. Together, these results suggest that biological systems employ quantum correlations in at least two distinct regimes — routing energy (FMO) and gating information (neural) — depending on the functional demand.

V. DATA AND CODE AVAILABILITY

The Lindblad solver is implemented in Python using NumPy/SciPy with Liouville-space matrix exponentiation. The FMO Hamiltonian, dephasing operators, and trapping terms follow the standard formulation [1, 3]. Source code is available at <https://github.com/dhira-jnitk/subnanometer-trp-transport>. A preprint of this manuscript is archived at Zenodo [6].

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